



Capacity estimation algorithm with a second-order differential voltage curve for Li-ion batteries with NMC cathodes



Taedong Goh ^a, Minjun Park ^b, Minhwan Seo ^b, Jun Gu Kim ^c, Sang Woo Kim ^{b,*}

^a Department of Creative-IT Engineering, Pohang University of Science and Technology, Pohang, 37673, South Korea

^b Department of Electrical Engineering, Pohang University of Science and Technology, Pohang, 37673, South Korea

^c Control System Research Department, POSCO ICT, Seongnam, 13486, South Korea

ARTICLE INFO

Article history:

Received 15 February 2017

Received in revised form

26 May 2017

Accepted 23 June 2017

Available online 24 June 2017

2010 MSC:

00-01

99-00

Keywords:

Differential voltage curve

Prominence peak

Curve alignment

Cycle aging

ABSTRACT

Accurate diagnosis of battery degradation is important for safe and efficient battery management. Capacity is a reliable index to describe the state of health (SOH) in batteries. In this paper, a capacity estimation algorithm for Li-ion batteries with nickel, manganese, and cobalt (NMC) cathodes based on a second-order differential voltage is proposed. A reference voltage curve was obtained during the CC charging phase from a fresh battery beforehand, and the input voltage curve was measured and compared, under the same operating conditions, from an aged battery. The input voltage curve is aligned to the reference curve to minimize the error of the second-order differential voltage. The compensated charging time of the aligned curve has a linear relation with the battery capacity until capacity reduction reaches 23.5%. From the linear model, the capacity can be estimated easily. This method is verified for five packs aged with different discharge currents. In the aging cycle and the initial SOC variation test, the capacity estimation error is less than 2% until it reaches 76.5% capacity. The proposed method does not require a complete aging test (for the table) to relate the charging time and the capacity.

© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

Li-ion batteries have been applied in various areas such as energy storage system and electric vehicle (EV) because of their high energy density and efficiency [1–4]. However, once a battery is installed, degradation of the battery begins whether it is used or not. A major reason for the degradation is the formation of a solid electrolyte interphase (SEI) [5,6]. It is created by reactions at the interface between anode and electrolyte. This phenomenon consumes available Li-ions in the electrolyte and prevents the flow of current. Older batteries generally have the lower capacity and higher internal resistance than fresh batteries, leading to a reduction of available energy and power [7–10]. Moreover, the growth of dendrites in a severely aged battery can cause internal short circuits that could initiate thermal runaway [11]. For all these reasons, diagnosis of the aging status is crucial for safe and efficient battery management. State of health (SOH), generally calculated by

impedance or capacity, is an indicator of the status of degradation for the battery [12,13].

The SOH estimation methods based on impedance have the advantages of shorter test time and a wider range of application, not limited by the depth of discharge (DOD). This is because small current variation is sufficient to obtain the impedance [14–16]. Pure resistance can be used as an aging indicator, and the electrochemical impedance test can also be utilized to analyze other elements, such as charge transfer resistance. Similarly, sample entropy, representing the degree of voltage fluctuation for the same current profile, has also been adopted to estimate SOH [17]. However, a drawback of this method is that the impedance may vary even with the same capacity fade [18]. The impedance alone cannot give direct information about the energy storage capability. For this, it is essential to determine the relation between resistance and capacity from prior aging test.

Battery capacity is a reliable index for describing aging status because it can be calculated by measuring current under controlled conditions. In such case, the SOH is assessed using the capacity ratio of the current battery to a fresh battery ($SOH = C_{current}/C_{fresh} \times 100\%$). However, the direct calculation method requires full discharge, which is inefficient from an energy

* Corresponding author.

E-mail addresses: ehd1116@postech.ac.kr (T. Goh), parkmj@postech.ac.kr (M. Park), mhseo09@postech.ac.kr (M. Seo), hopuler08@poscoict.com (J.G. Kim), swkim@postech.ac.kr (S.W. Kim).

perspective. There are many studies in which the battery capacity is estimated to provide a SOH index [19–34]. Model-based methods use equivalent battery models to obtain state of charge (SOC) and to calculate the capacity as the ratio of the time integral of the current, to the difference of the SOC for the same period ($C_{max} = \int_{t_1}^{t_2} Idt / (3600(SOC(t_2) - SOC(t_1)))$) [19–23]. These methods can be applied under various load current profiles, which means that invasive tests with fixed current profiles are not required. Nevertheless, excitation of the current is necessary to separate the open circuit voltage (OCV) and impedance parts. Moreover, periodic acquisition of an OCV–SOC table is necessary because the relation is changed by aging [35,36].

Learning based methods estimate the capacity using trained relations with various features extracted from the current and voltage profiles [24–31]. Those methods are appropriate for describing nonlinear relations between characteristic features and capacities and show accurate estimation results. In realistic test profile such as urban dynamometer driving schedule, internal resistance, polarization resistance, voltage change, and voltage were used to estimate the capacity through least squares support vector machine [29]. To eliminate additional test procedures, constant current (CC) and constant voltage (CV) charging profiles were used to extract the target features [24–27]. In addition, the initial charging SOC can be varied in practical applications; therefore, some of the research has been focused on extracting features by reducing the estimation error using restricted training data [25–27]. The feature of initial charging voltage, and charged capacities in CC and CV phases, were studied [25]. Characteristics only in the CV charging phase were also considered [26,27]. One limitation of all this previous research has been insufficient experimental validation for aged batteries using different initial charging SOC. The voltages of relaxed batteries after CC–CV charging was considered a feature insensitive to variation of initial SOC [28]. Incremental capacity analysis (ICA) has been conducted to determine the capacity under the narrow SOC range [30,31]. Nevertheless, the leaning-based algorithms require acquisition of data covering the entire age of the battery, which takes from several months to even years.

Recently, empirical methods based on the differential voltage curve have been proposed [32–34]. The voltage of battery at equilibrium status is determined by voltage difference between cathode and anode, and qualitative analysis of aging phenomenon for given the half cell voltages is possible [36–38]. Compared with the ICA, derivatives of voltage with respect to a charged capacity (dV/dQ) curve have advantage of easier analysis of separate electrodes because the voltage characteristic of both electrodes is kept linearly for acquisition of the derivatives [37,38]. The empirical methods are focused on nonlinear voltage characteristics insensitive to battery degradation, and the extracted features are based on the dV/dQ curve, which has a linear relationship with capacity especially. The first of these methods calculated the capacity based on a peak point, at high SOC on the dV/dQ curve [32]. This method could estimate the capacity under low DOD when the initial charging SOC was less than the peak at the highest SOC in the dV/dQ curve. Therefore, the total capacity could be calculated based on the current measured after the peak point at the highest SOC, without a low initial SOC condition. However, this method requires a low charging current to identify the second point and requires long testing time. The second method used charging time between two inflection points along the dV/dQ charging curve to estimate the capacity [33]. The time has a linear relation with capacity, but this method can only be applied to $LiFePO_4$ batteries, which exhibit an acute surge of voltage in high SOC range. Although the capacity estimation algorithms for batteries with nickel, manganese, and cobalt (NMC) were also proposed, they also still require a low

charging rate [34]. Consequently, the features used to estimate the capacity in the previous empirical methods are sensitive to the relatively high initial SOC and charging current.

In this paper, for a Li-ion battery with an NMC cathode, a capacity estimation algorithm using second-order differential voltage curve is proposed. The input charging curve is aligned based on a reference curve, to reduce error by the initial SOC. Moreover, the capacity of the input battery is estimated from the linear relation between the compensated charging time after alignment, and the capacity. Therefore, the capacity can be estimated from the reference charging voltage curve of a fresh battery, using a linear model of the relation between the compensated charging times and capacities. The proposed method shows accurate estimation results under a practical range of charging current.

In the rest of this paper, cycle aging experiment for battery packs with NMC cathodes and analysis of dV/dQ curve in cycle aging data is described in Section 2. In Section 3, a d^2V/dQ^2 based capacity estimation algorithm is explained. The alignment method of the input curve to compensate the charging time, selection of the reference curve, and a capacity estimation algorithm that uses the charging time are also discussed. Verification results for cycle aging data and initial SOC varying test are provided in Section 4. The validity of the model between capacity and the charging time is discussed in Section 5. Finally, the conclusions are presented in Section 6.

2. Aging data analysis

2.1. Aging test

Five battery packs were used to acquire the aging data. In each pack, two cells were coupled in parallel and the four groups were connected in series. Each cell in the packs was cylindrical and consisted of an intercalation graphite anode and lithiated metal oxide with cobalt, nickel, and manganese cathode (INR 18650-20R, SAMSUNG). The nominal capacity of the battery packs was 4 Ah, and voltage operation range was from 10 V to 16.8 V. The C-rate is determined by the nominal capacity, which defines 1C as 4 A for those packs. The battery packs were in a thermal chamber, and the ambient temperature was kept at 25 ± 2 °C. The batteries were charged and discharged using a battery test device (Regenerative Battery Pack Test System 17020, Chroma) with 5 s sample period. At first, the packs were cycled up to twenty times, and tests for capacity, direct current internal resistance (DCIR), and dynamic stress were performed in turn. The true capacities and CC charging voltage data were obtained in the capacity test, and internal resistances were calculated from the DCIR test. The cycle number of a set of the test cycles was set to twenty. In the aging cycle, the packs were charged in accord with the CC–CV protocol. The CC was 0.5C (2 A) with upper cutoff voltage 16.8 V, and cutoff current in CV phase was 0.05C (0.2 A). After charging, the constant discharge currents with five different values (1C, 2C, ..., and 5C) were applied to each pack until the lower cutoff voltage 10 V. Each battery pack is referred to as battery B1–B5, in the same order as the discharge currents. The capacity test consists of the same CC–CV charge phase with the aging cycle and 0.2C discharge until the 10 V cutoff voltage. True capacities were calculated using the accumulated current multiplied by the number of sampling periods in the discharge phase.

Charging voltage curves in the capacity test were used to estimate the capacity. The changes of the curve during the aging cycles are shown in Fig. 1, and some unreliable cycle data distorted for unidentified reasons were removed. Generally, for the cycled batteries, the voltage rose more rapidly, and the charging time in the CC phase decreased. These changes can be interpreted by capacity

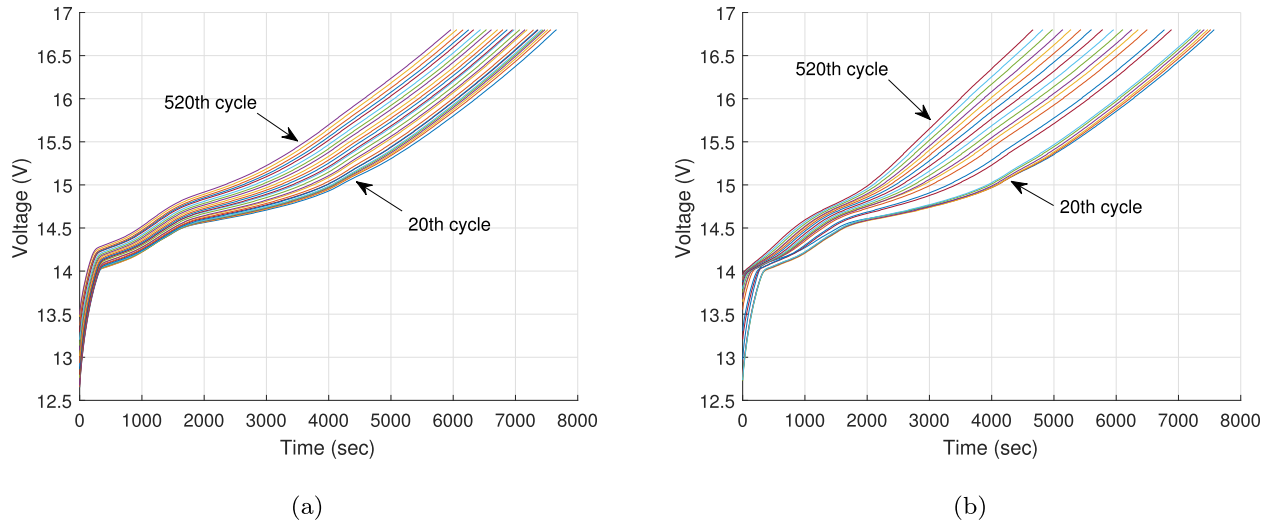


Fig. 1. Voltage curves for constant current charging along aging cycle: battery a) B1 and b) B5.

reduction and rise in impedance. In the aging cycles, the discharge current is constant; thus the voltage can reach the lower cutoff voltage earlier for higher impedance. In addition, the higher discharge current also causes a faster decrease in the voltage. Therefore, the initial voltage of the battery aged with 5C discharge current, shows a higher initial voltage and SOC, than does a battery aged at the 1C rate.

The aged batteries with five different discharge currents had different aging characteristics. For the higher current, capacity reductions were more severe for the same number of aging cycles (Fig. 2(a)). These results are consistent with those from other research [7,8]. However, internal resistance showed a quite interesting result in our experiment. In the DCIR test, periodic current pulses after 1 h rest were applied to SOC with about 10% intervals. When the pulse discharge current was applied at 90% SOC, the resistance was calculated by $R = (V(10) - V(0)) / (I(10) - I(0))$ where $V(10)$ and $I(10)$ is the voltage and current of the battery, measured 10 s after the pulse discharge. Impedances of the battery with a high discharge current decreased for the same capacity, as shown in

Fig. 2(b). This result was not deeply discussed in other research, but the reason can be guessed. It is likely due to cracks in the SEI and low DOD in the aging cycles [8,39,40]. For cycling with high current, the volume of the anode changes a great deal, which causes many cracks and dendrites in the SEI layer [39]. They enlarge the surface area of the layer, and the resistance becomes smaller for the same capacity fade. In addition, the battery was not fully discharged during the aging cycles with high discharge rates such as 5C, which delayed increase in the resistance [8,40]. This result shows that capacity may not be accurately estimated by impedance alone, without information about the aging condition such as C-rate and DOD.

2.2. Characteristics of the dV/dQ curve during the aging cycle

As discussed in Section 2.1, during the experiment, the initial SOC of the charging curve varies with the aging cycle and the current discharged. Variation of the initial SOC resulted in an unreliable index based on the CC charging time. In addition, fully

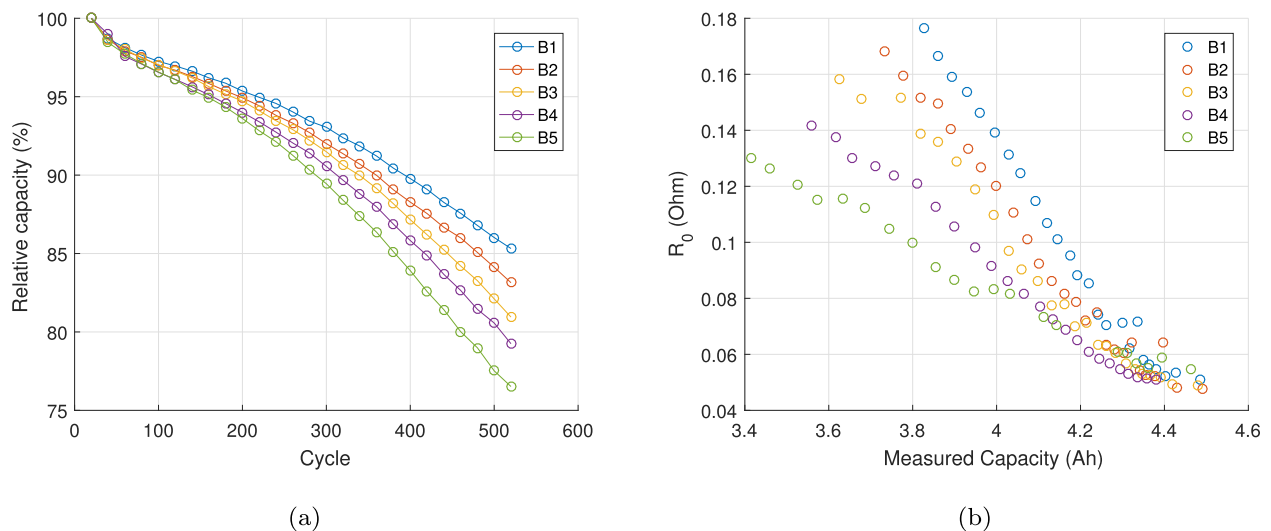


Fig. 2. Changes of the capacity and resistance for five batteries aged with different discharge current: a) relative capacity along aging cycle and b) relation between resistance and capacity.

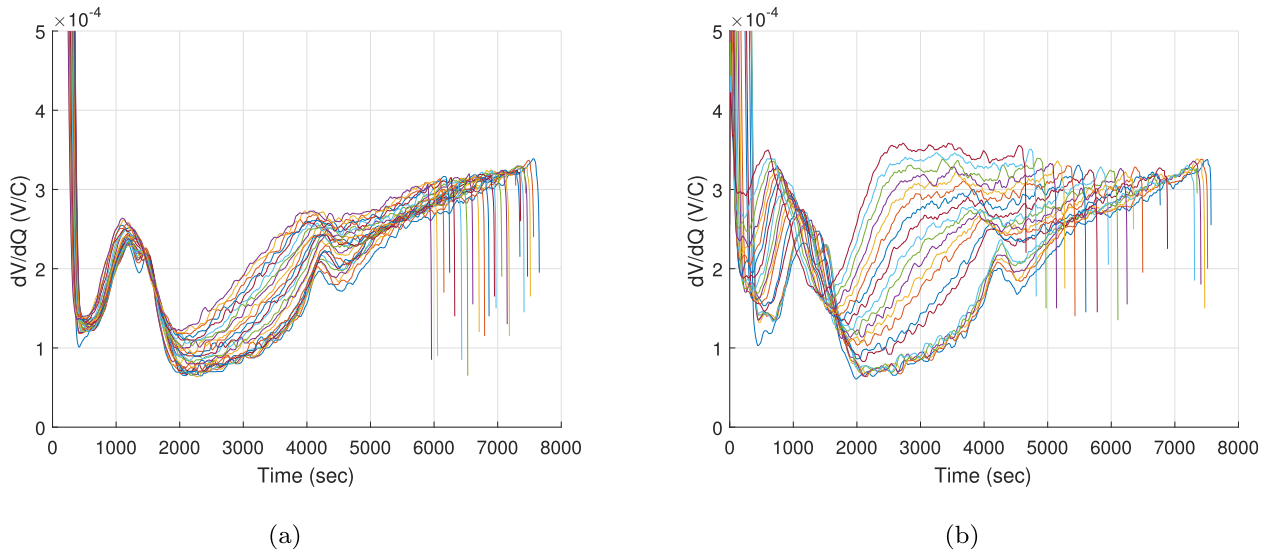


Fig. 3. dV/dQ curves under constant current phase along aging cycle: a) 1C discharged battery and b) 5C discharged battery.

discharging to make consistent initial SOC is a time-consuming process in real applications. Therefore, a feature insensitive to initial SOC on the dV/dQ curve of charging voltage is required. Here, dV/dQ can be calculated using $dV/dQ = (V(t+T) - V(t)) / ((I(t+T) - I(t))T)$, where t is current time and T is sample period. However, this causes a large fluctuation due to measurement noise. To eliminate it, a median filter and moving average filter were applied in turn. The intermittent impulse noise was removed by the median filter, and continuous and low-level noise was reduced by the moving average filter. The results are shown in Fig. 3(a) and (b). Rises in dV/dQ during the aging cycles for the five battery packs were observed. In Fig. 3(a), there are two peaks on the dV/dQ for the all aging cycles. The first and second peaks are revealed in the low and middle SOC range, respectively. Despite degradation, the shape of dV/dQ near the first peak was almost unchanged, and only a small offset was added. However, the curve at the second peak changed drastically. The peaks were even difficult to distinguish in severely aged batteries (see Fig. 3(b)). Compared with results from previous papers [32,34], the second peak was not appropriate as a

characteristic point in our experimental data because the voltage fluctuation was not negligible when charging with high current (0.5C). The previous workers also insist that low current (about 0.2C) was necessary to identify the second peak. Therefore, the first peak was selected as the characteristic point because it was insensitive to the current amplitude.

The initial SOC of the acquired charging curve are inconsistent, thus the relation between partial charging times, based on the characteristic point and capacities, were analyzed. The partial charging time $t_{CC, peak based}$ is defined as the difference between the time at the first peak and at the end of the CC charging phase. Although filtering and smoothing methods were applied, the dV/dQ curves still had many local maximum points that could result in errors during peak detection. To find the peak accurately, the prominence peak detection method was applied. The prominence peak detection algorithm detects the dominant peak points based on selected prominence and width. The prominence disregards unimportant peaks near the main first peak using a dV/dQ threshold determined by dominance, which is based on

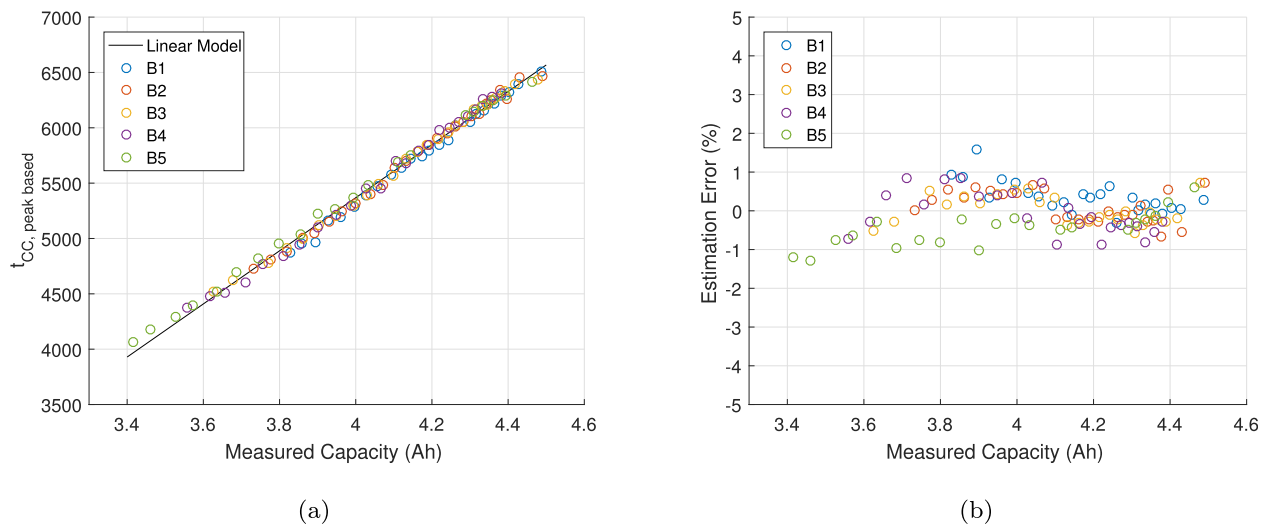


Fig. 4. a) Relation between extracted charging time and capacity and b) estimation error.

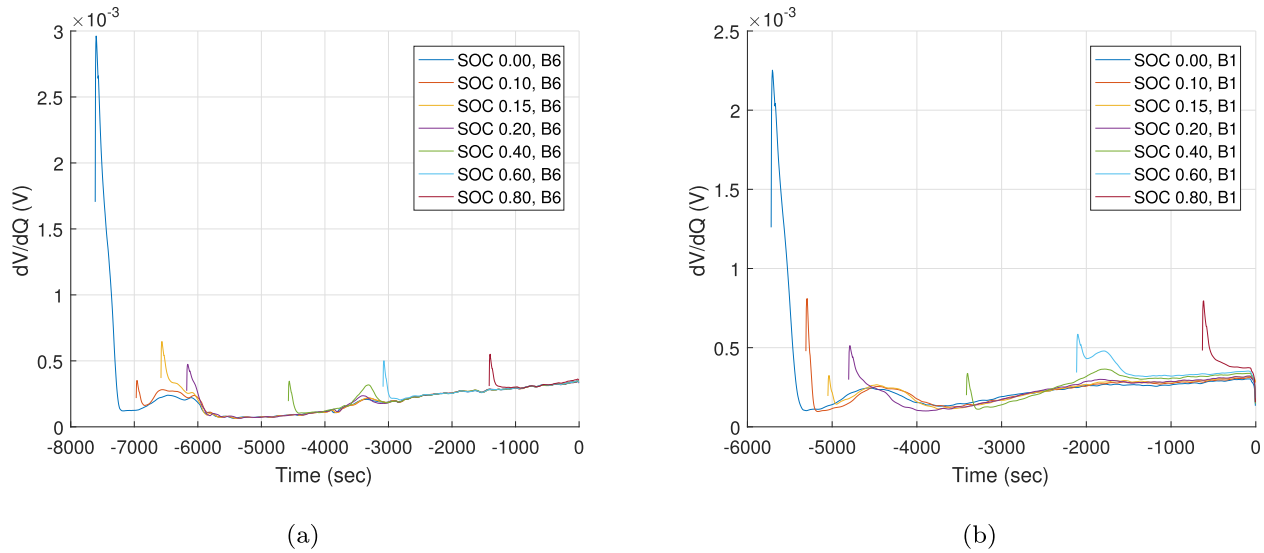


Fig. 5. dV/dQ curves with different initial SOC under CC charging phase: a) fresh battery (97.8% SOH) and b) aged battery (80.0% SOH).

surrounding local minimum points. The width factor removes insignificant peak points by restricting the width of the peak. The most remarkable prominence value of dV/dQ for all aging cycles is calculated by the difference between the values of the first downward point and the first peak. The method was implemented simply using MATLAB toolbox [41], for which MinPeakProminence and MinPeakWidth were set as 5×10^{-5} V/C and 30 s respectively. The $t_{cc,peak\ based}$ is calculated from the interval between the extracted peak point and the end of the CC charging phase, and its relation with the capacity is shown in Fig. 4(a). Note that the relation was linear and similar to the five batteries with different discharge currents. For the cycle aging data, the peak detection method is sufficient to estimate the total capacity using the linear relation.

2.3. The dV/dQ curve characteristic for initial SOC

To consider changes of the initial SOC in real applications, the peak detection method still needs to be verified. The dV/dQ curves of a fresh battery with 97.8% SOH, and of an aged battery with 80.0% SOH, at the seven initial SOC (0, 10, 15, 20, 40, 60, and 80%) are compared in Fig. 5(a) and (b). The charging curves were acquired after every individual test with CC-CV charging the same as for the aging cycle, 0.5C CC discharging until the initial SOC, and 30 min rest. The curves were aligned on the time axis from the right side, based on the end point of the CC phase, to compare the location of the first upward peak. The peak occurred when the initial SOC was

below 10% for the fresh battery, and 15% for the aged battery. Therefore, initial SOC above 20% is not appropriate for identifying the first peak. It is noteworthy that dV/dQ of the aged battery did not converge to the same curve, even after adequate relaxation time, for different initial SOC. The effect of the initial SOC on the following charging curve, including the second peak, may not be negligible for aged batteries.

The initial SOC of the aged pack affected the following shape of dV/dQ more than the fresh pack did. The effect of low SOC (below 10%) for both packs was less significant on the first peak, which means that the first peak could be used for the CC curve with low SOC to determine the capacity. However, the alignment could be disturbed by distortion of dV/dQ when the initial SOC was near the peak, and the peak could even be eliminated.

3. Proposed algorithm

The proposed algorithm compensates the CC charging time using a d^2V/dQ^2 curve segment although the initial SOC is not zero. Based on our experimental data, the sign of the dV/dQ slope is divided into four parts (Fig. 6). On the left side of the first peak, there are two intervals with different signs, and they are sensitive to even low initial SOC. In contrast to the peak detection method, the charging curve is aligned by similarity with a reference curve using a d^2V/dQ^2 segment, which is the downhill curve of d^2V/dQ^2 after the first peak. This method can be applied to a voltage curve starting from higher SOC with ambiguous dV/dQ peaks because the method determines the similarity using more points on the downhill slope.

The steps of the proposed algorithm are characteristic curve acquisition from the input voltage, CC charging time estimation using curve alignment, capacity estimation, and reference update (Fig. 7). At first, the characteristic curve is extracted from the dV/dQ of the input voltage curve in CC charging phase. The differentiation is the same step as in Section 2.2, which includes filtering and smoothing. The characteristic curve means the downhill curve after the first peak, as shown in Fig. 6. The characteristic curve is obtained as the partial curve between the first peak (t_1) and second valley (t_2), and a sign of the slope of the curve segment is negative. The second-order differentiation is also calculated by differentiating dV/dQ again. Here, d^2V/dQ^2 is used to match the same

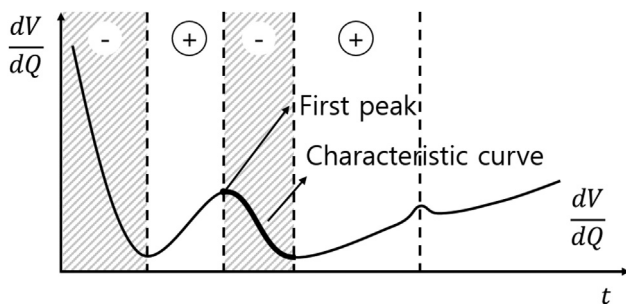


Fig. 6. Signs of slope in dV/dQ curve.

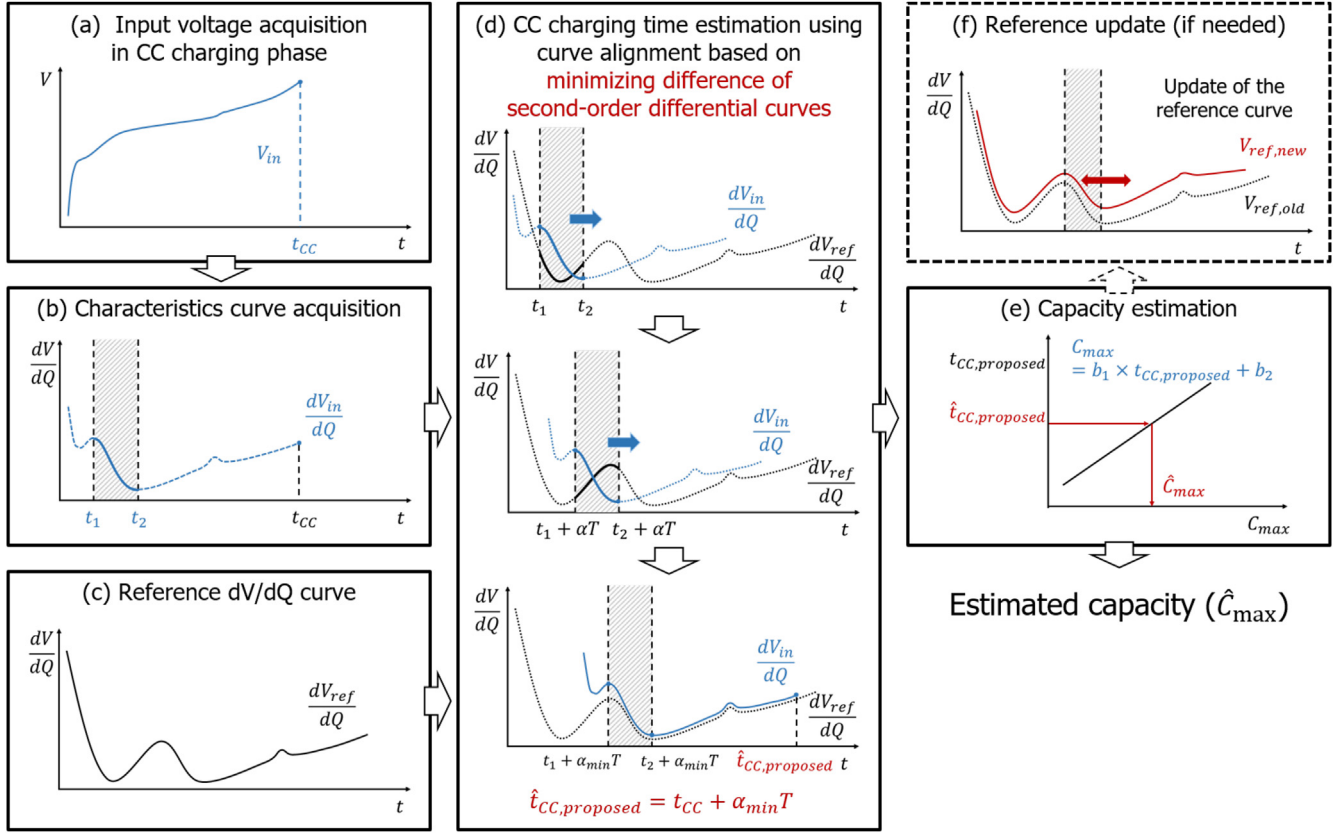


Fig. 7. Flowchart of the proposed algorithm.

characteristic curve. The curve alignment is proceeded by matching the input voltage curve to the reference curve in time axis based on closeness of the characteristic curves. For clarity, only dV/dQ is depicted, but the alignment of an input charging curve using the similarity of the dV/dQ could cause significant error because of the rise in dV/dQ caused by battery degradation. On the other hand, the characteristic curve still has negative values for aged batteries, and the signs of the surrounding regions of the curve are positive. Therefore, the sign of the curve is an appropriate criterion of the curve alignment. In addition, the characteristic curve is less sensitive to initial SOC. Consequently, the curve alignment problem can be formulated as in the following optimization equation with second-order differentiation as:

$$\alpha_{min} = \arg \min_{\alpha} \sum_{k=k_1+\alpha}^{k_2+\alpha} \left(\frac{d^2 V_{ref}(kT)}{dQ^2} - \frac{d^2 V_{in}((k-\alpha)T)}{dQ^2} \right)^2, \quad (1)$$

where V_{ref} is the reference voltage, V_{in} is the input voltage, k_1 is the first upward peak point ($t_1 = k_1 T$) in discrete time, and k_2 is the second downward peak point ($t_2 = k_2 T$) in discrete time. The optimization problem in equation (1) finds an interval (αT) minimizing the sum of squared errors between the characteristic curve shifted by αT ($d^2 V_{in}/dQ^2|_{t \in [t_1, t_2]}$) and the reference curve ($d^2 V_{ref}/dQ^2|_{t \in [t_1+\alpha T, t_2+\alpha T]}$) corresponding to time axis. After the alignment, the signs of the two dV/dQ slopes meet each other as in Fig. 7. In equation (1), $\alpha_{min} T$ indicates compensated time of t_1 . After solving the problem, the compensated CC charging time can be represented as:

$$t_{CC,proposed} = t_{CC} + \alpha_{min} T, \quad (2)$$

where t_{CC} is the total charging time in the CC phase for the V_{in} .

The relation between the compensated charging time and the total capacity has not been verified theoretically. Therefore, an empirical model-based on data from the aging test was used in the capacity estimation step, and the relation is assumed to be linear as:

$$C_{max} = b_1 \times t_{CC,proposed} + b_2, \quad (3)$$

where C_{max} is the total capacity of the battery, and b_1 and b_2 are parameters of the model. Equation (3) will be discussed in the following (Results) section. To estimate parameters of the linear model, at least two different data sets, including the compensated charging time and the capacity, are required. Here, b_1 and b_2 are estimated using a least-squares algorithm [42]. To determine the capacities, the CC-CV charging (0.5C CC charging up to 16.8 V and CV charging until 0.05C) and the 0.2C discharge tests are required. However, once the model is obtained, the capacity test is unnecessary. In addition, the linearity guarantees that the capacity test can be conducted for any degradation status that satisfies the linearity of the model. In other words, a preliminary aging test using a sample battery is not required as with learning-based methods.

The reference curve in the proposed algorithm is directly related to the estimation accuracy. Basically, the reference curve can be obtained from the charging voltage curve of the fresh battery under the same CC phase as the input curve acquisition after 0.5C full discharging. However, the shape of dV/dQ near the first peak point curve changes during the aging cycle, and the drastic deformation

can cause a large error. Therefore, alteration of the reference curve is required for deeply aged batteries. For consistent charging time acquisition, a new reference curve was aligned based on the old reference curve using equation (1), and the time index of the new curve was also shifted. After alignment of the new reference, the charging time of the input voltage curve was estimated from the new reference curve.

4. Results

4.1. Estimation result for the aging cycle test

For verification of the proposed algorithm, the same voltage curves of the differently aged five batteries were used as in Section 2.1. To compare overall accuracy, the parameters of equation (3) were acquired from the whole cycle aging data ($b_1 = 0.4169 \times 10^{-3} \text{ Ah} \cdot \text{s}^{-1}$, and $b_2 = 1.2726 \text{ Ah}$). At first, the reference charging curve was obtained from the battery B1 at the 20th aging cycle, and the reference update method was not applied. After implementing the curve-alignment algorithm, most of the first peaks for the cycle data were aligned at same location, as

shown in Fig. 8(a) and (b). The charging time has a linear relation with the capacity as in Fig. 8(c), and this result is clearer than for the peak based algorithm because the proposed method is less affected by peak detection error. The capacity estimation error is calculated by $\text{error} = 100 \times (C_{\max} - C_{\max, \text{est}}) / C_{\max} \%$, where $C_{\max, \text{est}}$ is the estimated capacity. The magnitude of error is less than 2% without a voltage data set. In the capacity estimation result, the data from the 520th aging cycle of battery B5, which was the most severely aged in the aging cycle test, shows large estimation error (about 5%). This phenomenon occurs because of variation in the nonlinear charging curve with degradation. The voltage curve at the 520th cycle of battery B5 did not match accurately the data of battery B1 at the 20th cycle. Therefore, the reference curve update method was applied after some capacity degradation. Among the experimental data, the 500th aging cycle test data was selected for use as the new reference curve. The estimation result is also plotted in Fig. 8, the error of the 5th battery at the 520th aging cycle was reduced as much as 1.3799%.

The capacity estimation result of the proposed method for each battery pack was compared with the peak based algorithm. In the case of the peak based method, the partial charging time between

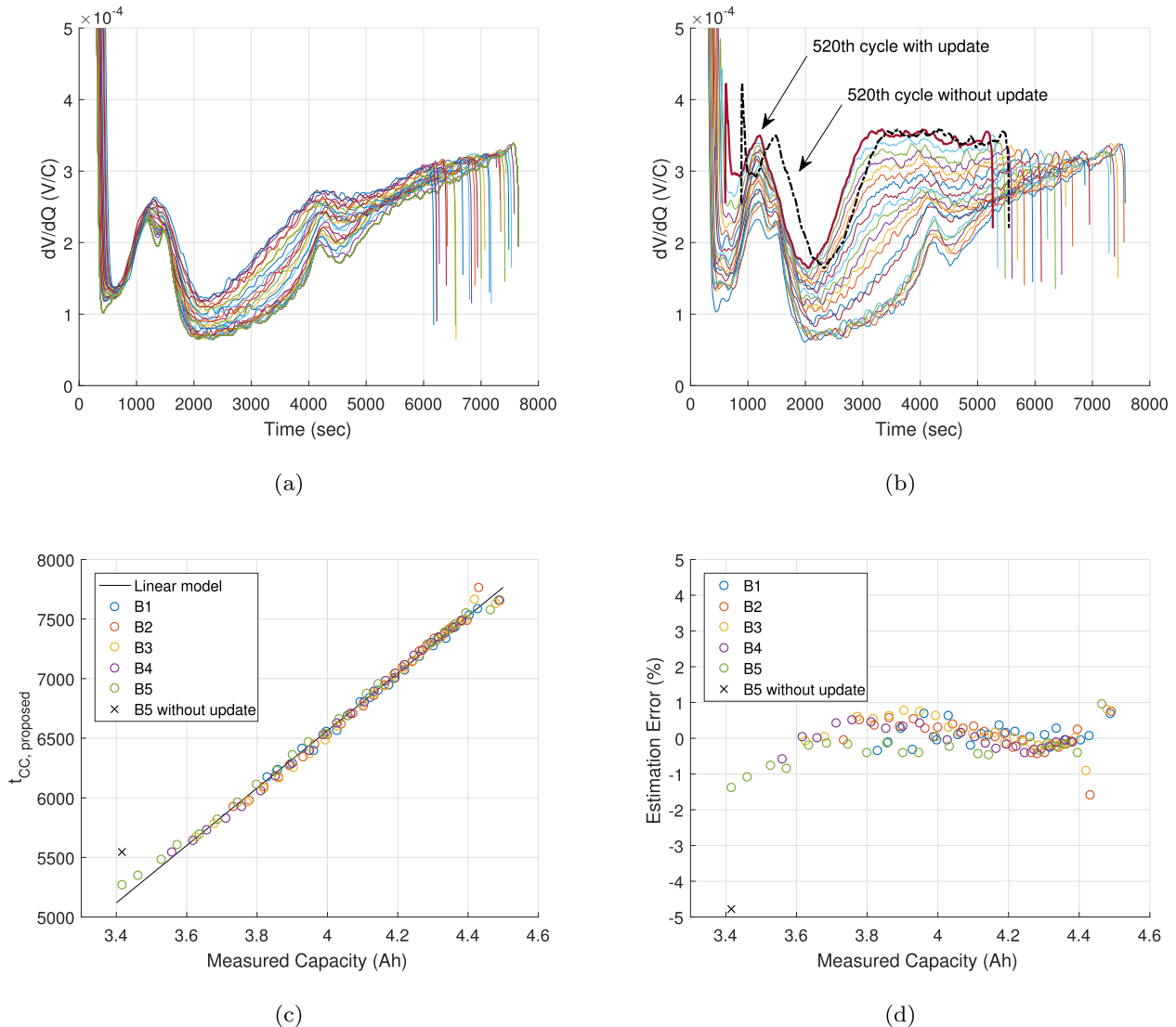


Fig. 8. Aligned charging curve and estimation result of the proposed method with the reference update method: a) B1, b) B5, c) relation between the compensated charging time and capacity and d) estimation error.

Table 1
Capacity estimation error of the peak based algorithm for cycle aging data.

	B1	B2	B3	B4	B5	total
RMSE (%)	0.5343	0.4071	0.3658	0.5467	0.6536	0.5056
Maximum error (%)	1.5883	0.7263	0.7241	0.8845	1.2865	1.5883

Table 2
Capacity estimation error of the proposed algorithm for cycle aging data.

	B1	B2	B3	B4	B5	total
RMSE (%)	0.2922	0.4639	0.4215	0.2900	0.5646	0.4154
Maximum error (%)	0.6947	1.5837	0.9041	0.5843	1.3799	1.5837

the first upper peak and the end of the CC charging phase was used for t_{CC} in equation (3), and the estimation result is shown in Fig. 4(b) and Table 1 as the root mean square error (RMSE) and maximum error. The errors of the proposed algorithm are represented in Table 2. The maximum error of the proposed method was 1.5837%, and this is similar to that of the peak based algorithm. The RMSE of the proposed method was 0.4154%, which is lower than that of the peak based (0.09%). To analyze the estimation errors further, histograms of both methods are compared in Fig. 9. The shapes of approximated density functions are similar, but the number of samples near to zero is larger for the proposed algorithm. Therefore, the proposed method is more accurate than the peak based capacity estimation method. In addition, the differences in the estimation errors of both algorithms from battery B1 to B5 were insignificant. As mentioned in Section 2.1, the impedance of each battery is different, which means that the proposed algorithm is insensitive to impedance change from aging paths with different discharge rates. However, variation in the pure resistance, caused by external factors, could affect the compensated charging time.

An important part of the reference update method is the time when the reference curve is updated. In our data set, the significant error was not revealed until up to 76.5% SOH. However, this could be different for other battery types. Therefore, periodic change within 23.5% SOH reduction of the reference curve may be possible. On the other hand, too frequent updating of the curve could cause accumulated capacity error from curve alignment. For this reason, too frequent updating should be avoided.

4.2. Estimation result for SOC varying test

The initial SOC affects the following charging curve, especially for aged batteries. In addition, the first peak does not appear in dV/dQ for high initial SOC ($> 20\%$). The capacity estimation algorithm was verified for low initial SOC ($< 15\%$). The SOC varying test was conducted for five aged batteries rested for several months after cycle aging, and a fresh battery was referred to as battery B6. In the SOC varying test, the packs were stored without controlling temperature. The initial SOC was controlled in relation to the capacity measurement, the full charge with the CC-CV charging protocol and partial discharge to the SOC. The three initial SOC ($0, 10$, and 15%) were used to compare the estimation accuracy. After alignment, the first peaks in the CC charging appear at the same moment for the aged battery (Fig. 10(a)), which can be considered as having the same remaining capacity. The SOC corresponding to the peak was higher for older batteries because the SOC is calculated by the ratio of the remained capacity to the total capacity. The actual capacities were obtained from a following 0.2C discharging test after the CC-CV charging from the initial SOC, to eliminate error from the cycle aging. Here, 0% means that the battery is fully discharged until the cutoff voltage with CC, which means that the initial SOC was not exactly 0%. For the fresh pack at 15% initial SOC, the first peak does not appear clearly, thus, in this case the first peak point of the proposed method is replaced by the local maximum point of d^2V/dQ^2 until the second valley point. To prevent selecting the local maximum point in low dV/dQ region near to the second valley, the dV/dQ corresponding to the point is restricted to above 1.9×10^{-4} V/C. The dV/dQ s for different initial SOC of the fresh battery had similar downward slopes before the second valley as shown in Fig. 10(b) even if the first peak is not detected.

The estimation results of the peak based method and the proposed algorithm for SOC varying test are compared in Fig. 11. However, the data on estimated capacity for deeply aged batteries ($< 76.5\%$ SOH) were not linear; thus only the voltage data for battery B1, B2, and B6 were used for verification. The SOC error is less than 2% for the two aged batteries and the fresh battery, and the error in relation to the initial SOC was not significant. Therefore, the compensated charging time is a feature insensitive to the low initial SOC under 15%. On the other hand, the peak based algorithm could not detect most of the first peaks of dV/dQ when initial SOC is

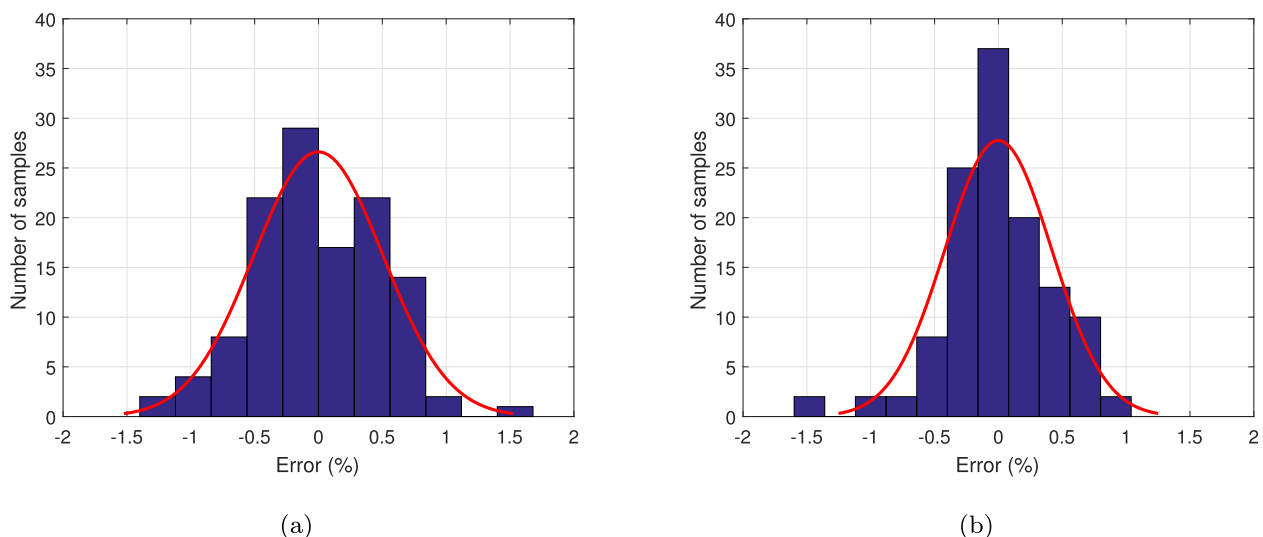


Fig. 9. Aligned charging curve and estimation result of the proposed method with the reference update method: a) B1, b) B5, c) relation between the compensated charging time and capacity and d) estimation error.

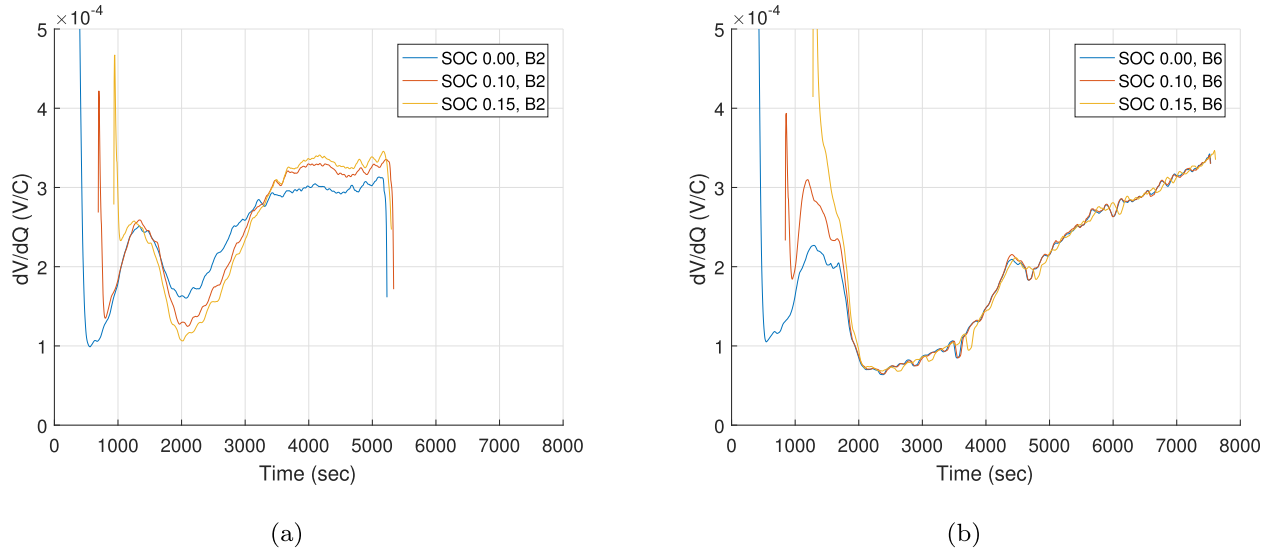


Fig. 10. dV/dQ curve with three initial SOC for the different batter packs under CC charging phase: a) B2 and b) B6.

15%. The algorithm also could not work for a case of 10% SOC. The peak based algorithm is implemented to find a peak point which is large enough to discriminate with small local peaks, so the algorithm could not be applied when the initial SOC is large as much as 15%. In addition, the capacity estimation errors of the proposed algorithm are lower than the peak based algorithm. The differences of the errors are small, but some errors for the peak based algorithm are out of the 2% error bound. The peak based algorithm has an advantage of wide capacity range under large error bound such as 4%, but could not be applied to batteries at 15% initial SOC. Consequently, the proposed algorithm is more accurate for capacity above 76.5% SOH and can be applied for higher initial SOC than the peak based algorithm.

5. Discussion

The relation between the capacity and the compensated charging time was assumed to be the first order polynomial in this work. However, theoretical verification of the linear model is

difficult. To the best of our knowledge, the linearity of the model can be shown using experimental results. The input and output of the model are charging time in the CC phase and the discharged capacity. The efficiency of the charging current is assumed to be 1 because the current was low as 0.5C and the batteries were charged with lower current in the CV protocol than the CC phase. Therefore, the total discharged capacity (C_{max}) can be represented as follows:

$$C_{max} = Q_{CC} + Q_{CV}, \quad (4)$$

where Q_{CC} and Q_{CV} are the charged capacities in the CC and the CV phases. Then, the linearity is can be explained by the two assumptions:

$$Q_{CC} = \frac{I_{CC} t_{CC, proposed}}{3600}, \quad (5)$$

$$Q_{CV} = a_1 C_{max} + a_2, \quad (6)$$

where I_{CC} is the magnitude of the current in the CC phase,

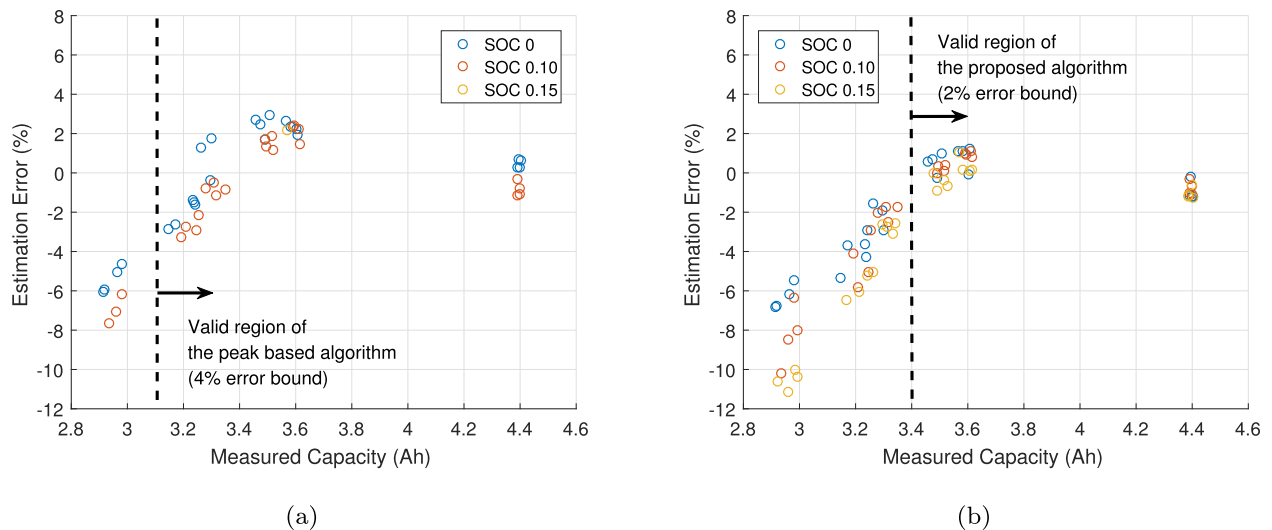


Fig. 11. Result of the proposed algorithm in SOC varying test: a) relation between the compensated charging time and capacity and b) estimation error.

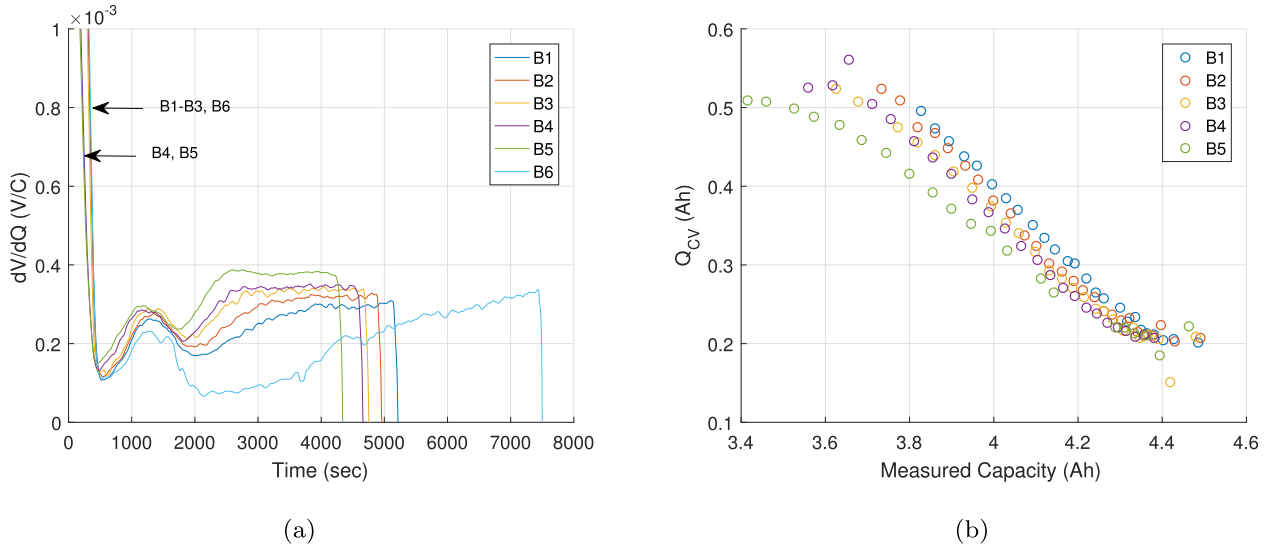


Fig. 12. a) dV/dQ curves for the same initial SOC, b) Charged capacities in CV the phase.

$t_{CC, proposed}$ is the compensated charging time by the proposed algorithm and a_1 , a_2 are constant parameters. If those assumptions are true, then the total capacity can be described as follows:

$$C_{max} = \frac{I_{CC} t_{CC, proposed}}{(1 - a_1) 3600} + \frac{a_2}{1 - a_1} = b_1 t_{CC, proposed} + b_2, \quad (7)$$

The estimated CC time can be equal to the actual time when the dV/dQ characteristics for low SOC is consistent in relation to degradation. The open circuit voltage of batteries can be affected by loss of active material in anode and cathode and loss of Li-ion [36]. This phenomenon is also applied in dV/dQ [37,38]. In this case, the first peak indicates phase transition voltage in anode. Based on the previous research, the most probable case affecting severely the dV/dQ in low SOC is active material loss in the cathode without Li-ion loss. If the situation is worsening, then dV/dQ before the first peak can be distorted. However, for 20% loss of capacity, the perfect conservation of Li-ion is unrealistic. For the fresh and aged batteries in the SOC varying test, dV/dQ acquisition test starting from completely discharged status was conducted. The batteries were

discharged at low current (0.05C) up to lower cutoff voltage (10 V) and then were charged in CC phase. The dV/dQ curves in the charging phase show that the dV/dQ curves prior to the first peak are similar for different capacities except for the battery B4 and B5 in Fig. 12(a). They show different characteristics of dV/dQ for initial SOC, which can be explained as an unbalanced loss of active materials.

The charged capacities in CV are presented in Fig. 12(b), and they are not exactly the capacities entered in the batteries because of the passive balancing circuit in the packs. However, overall tendency shows that the charged capacity in CV phase can be represented as first-order polynomial to the total capacity as in (6). For low capacities about 3.4 Ah, the Q_{CV} can be apart from the model, but the difference can be negligible because of the low portion of the Q_{CV} in entire charged quantity. For capacity below 3.4 Ah as in the SOC varying test, the large errors can be interpreted by the error Q_{CV} and distortion of dV/dQ .

One limitation of our algorithm is the large DOD for the charging phase. For some applications, the DOD can be restricted to improve cycle life. To check the feasibility of reducing DOD by restricting

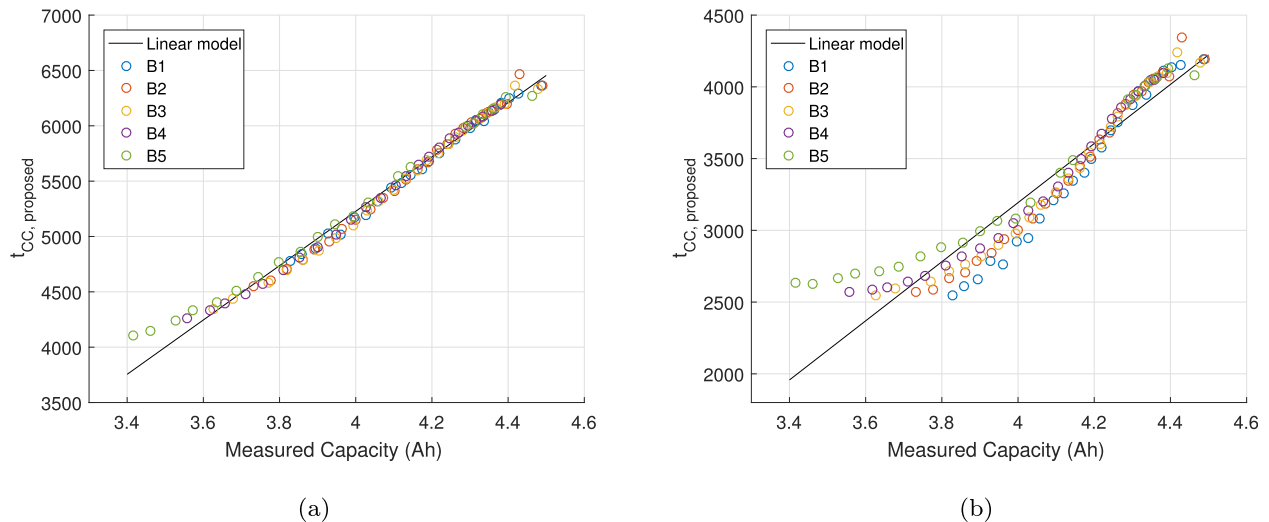


Fig. 13. Relation between charging time and capacity by different cutoff charging voltage: a) 16 V and d) 15 V.

upper SOC, variation in charging time due to the upper different cutoff voltage was analyzed. The low initial SOC is also a restriction of our algorithm, thus the lower SOC limit was not considered. The upper voltage limit does not make the upper SOC the same, but it can easily be implemented. The relation between the compensated charging time and the capacity under two different cutoff voltages (16 V and 15 V) for the cycle aging data were compared, and the relations are shown in Fig. 13. It is remarkable that the range of the capacity satisfying the linearity is reduced for the lower cutoff voltage. In addition, the charging times at the same capacity are different in relation to discharging current. In such cases, a learning-based algorithm considering more features related to capacities can be used to estimate the capacity. Therefore, regarding charging, the linearity and irrelevance of different aging paths are satisfied for high cutoff voltage.

6. Conclusions

An algorithm for better estimating the capacity of Li-ion batteries with NMC cathodes was proposed in this paper. The charging voltage curve in CC phase was used to acquire the characteristic curve including the first peak and was aligned based on the second-order derivative of the voltage curve. Five battery packs cycled with different discharge currents were used to verify the accuracy of the proposed algorithm. The relation between the compensated charging time and the capacity was linear up to 23.5% SOH degradation and insensitive to the aging cycle with a different discharging C-rate. The capacity was estimated using a linear model, and its error was less than 2%. In addition, the estimation error was comparable, for different initial SOC's below 15%, and for aged batteries as well. This algorithm can be applied without a prior aging experiment to construct a lookup table, and only several tests, including charging voltage acquisition and capacity measurement, are required. Therefore, the proposed algorithm can reduce the laborious experiments required previously, and the accuracy of the capacity estimation is acceptable. One disadvantage of the new algorithm is that the initial SOC is low (about 15%). Verification of the algorithm for different charging current and cathode types, and prediction of the remaining useful life using the compensated charging time will be the focus of future work.

Acknowledgment

This research was supported by the MSIP (Ministry of Science, ICT and Future Planning), Korea, Under the "ICT Consilience Creative Program" (IITP-R0346-16-1007) supervised by the IITP (Institute for Information & communications Technology Promotion).

References

- [1] Lukic SM, Cao J, Bansal RC, Rodriguez F, Emadi A. Energy storage systems for automotive applications. *IEEE Trans Ind Electron* 2008;55(6):2258–67.
- [2] Divya K, Østergaard J. Battery energy storage technology for power systems an overview. *Electr Power Syst Res* 2009;79(4):511–20.
- [3] Lawder MT, Suthar B, Northrop PV, De S, Hoff CM, Leiternann O, et al. Battery energy storage system (bess) and battery management system (bms) for grid-scale applications. *Proc IEEE* 2014;102(6):1014–30.
- [4] Diouf B, Pode R. Potential of lithium-ion batteries in renewable energy. *Renew Energy* 2015;76:375–80.
- [5] Verma P, Maire P, Novák P. A review of the features and analyses of the solid electrolyte interphase in li-ion batteries. *Electrochim Acta* 2010;55(22):6332–41.
- [6] He Y-B, Li B, Liu M, Zhang C, Lv W, Yang C, et al. Gassing in li4ti5o12-based batteries and its remedy. *Sci Rep* 2012;2:913.
- [7] Ning G, Haran B, Popov BN. Capacity fade study of lithium-ion batteries cycled at high discharge rates. *J Power Sources* 2003;117(1):160–9.
- [8] Schuster SF, Bach T, Fleder E, Müller J, Brand M, Sextl G, et al. Nonlinear aging characteristics of lithium-ion cells under different operational conditions. *J Energy Storage* 2015;1:44–53.
- [9] Krieger EM, Cannarella J, Arnold CB. A comparison of lead-acid and lithium-based battery behavior and capacity fade in off-grid renewable charging applications. *Energy* 2013;60:492–500.
- [10] Eddahech A, Briat O, Vinassa J-M. Performance comparison of four lithium-ion battery technologies under calendar aging. *Energy* 2015;84:542–50.
- [11] Harry KJ, Hallinan DT, Parkinson DY, MacDowell AA, Balsara NP. Detection of subsurface structures underneath dendrites formed on cycled lithium metal electrodes. *Nat Mater* 2014;13(1):69–73.
- [12] Waag W, Fleischer C, Sauer DU. Critical review of the methods for monitoring of lithium-ion batteries in electric and hybrid vehicles. *J Power Sources* 2014;258:321–39.
- [13] Bercibar M, Gandiaga I, Villarreal I, Omar N, Van Mierlo J, Van den Bossche P. Critical review of state of health estimation methods of li-ion batteries for real applications. *Renew Sustain Energy Rev* 2016;56:572–87.
- [14] Remmlinger J, Buchholz M, Meiler M, Bernreuter P, Dietmayer K. State-of-health monitoring of lithium-ion batteries in electric vehicles by on-board internal resistance estimation. *J Power Sources* 2011;196(12):5357–63.
- [15] Galeotti M, Cinà L, Giammanco C, Cordiner S, Di Carlo A. Performance analysis and soh (state of health) evaluation of lithium polymer batteries through electrochemical impedance spectroscopy. *Energy* 2015;89:678–86.
- [16] Yuan H-F, Dung L-R. Offline state-of-health estimation for high-power lithium-ion batteries using three-point impedance extraction method. *IEEE Trans Veh Technol* 2017;66(3):2019–32.
- [17] Hu X, Li SE, Jia Z, Egardt B. Enhanced sample entropy-based health management of li-ion battery for electrified vehicles. *Energy* 2014;64:953–60.
- [18] Schuster SF, Brand MJ, Campestrini C, Gleissenberger M, Jossen A. Correlation between capacity and impedance of lithium-ion cells during calendar and cycle life. *J Power Sources* 2016;305:191–9.
- [19] Hu C, Youn BD, Chung J. A multiscale framework with extended kalman filter for lithium-ion battery soc and capacity estimation. *Appl Energy* 2012;92:694–704.
- [20] Xiong R, Sun F, Chen Z, He H. A data-driven multi-scale extended kalman filtering based parameter and state estimation approach of lithium-ion polymer battery in electric vehicles. *Appl Energy* 2014;113:463–76.
- [21] Kim T, Wang Y, Sahinoglu Z, Wada T, Hara S, Qiao W. A rayleigh quotient-based recursive total-least-squares online maximum capacity estimation for lithium-ion batteries. *IEEE Trans Energy Convers* 2015;30(3):842–51.
- [22] Zou Y, Hu X, Ma H, Li SE. Combined state of charge and state of health estimation over lithium-ion battery cell cycle lifespan for electric vehicles. *J Power Sources* 2015;273:793–803.
- [23] Wei Z, Zhao J, Ji D, Tseng KJ. A multi-timescale estimator for battery state of charge and capacity dual estimation based on an online identified model. *Appl Energy* 2017. <http://dx.doi.org/10.1016/j.apenergy.2017.02.016>.
- [24] Wu J, Wang Y, Zhang X, Chen Z. A novel state of health estimation method of li-ion battery using group method of data handling. *J Power Sources* 2016;327:457–64.
- [25] Hu C, Jain G, Schmidt C, Strief C, Sullivan M. Online estimation of lithium-ion battery capacity using sparse bayesian learning. *J Power Sources* 2015;289:105–13.
- [26] Eddahech A, Briat O, Vinassa J-M. Determination of lithium-ion battery state-of-health based on constant-voltage charge phase. *J Power Sources* 2014;258:218–27.
- [27] Lu C, Tao L, Fan H. Li-ion battery capacity estimation: a geometrical approach. *J Power Sources* 2014;261:141–7.
- [28] Baghdadi I, Briat O, Gyan P, Vinassa JM. State of health assessment for lithium batteries based on voltage–time relaxation measure. *Electrochim Acta* 2016;194:461–72.
- [29] Deng Z, Yang L, Cai Y, Deng H, Sun L. Online available capacity prediction and state of charge estimation based on advanced data-driven algorithms for lithium iron phosphate battery. *Energy* 2016;112:469–80.
- [30] Weng C, Cui Y, Sun J, Peng H. On-board state of health monitoring of lithium-ion batteries using incremental capacity analysis with support vector regression. *J Power Sources* 2013;235:36–44.
- [31] Weng C, Feng X, Sun J, Peng H. State-of-health monitoring of lithium-ion battery modules and packs via incremental capacity peak tracking. *Appl Energy* 2016;180:360–8.
- [32] Bercibar M, Garmendia M, Gandiaga I, Crego J, Villarreal I. State of health estimation algorithm of lifepo 4 battery packs based on differential voltage curves for battery management system application. *Energy* 2016;103:784–96.
- [33] Wang L, Pan C, Liu L, Cheng Y, Zhao X. On-board state of health estimation of lifepo 4 battery pack through differential voltage analysis. *Appl Energy* 2016;168:465–72.
- [34] Bercibar M, Devriendt F, Dubarry M, Villarreal I, Omar N, Verbeke W, et al. Online state of health estimation on nmc cells based on predictive analytics. *J Power Sources* 2016;320:239–50.
- [35] Lavigne L, Sabatier J, Francisco JM, Guillemard F, Noury A. Lithium-ion open circuit voltage (ocv) curve modelling and its ageing adjustment. *J Power Sources* 2016;324:694–703.
- [36] Birkel CR, Roberts MR, McTurk E, Bruce PG, Howey DA. Degradation diagnostics for lithium ion cells. *J Power Sources* 2017;341:373–86.
- [37] Bloom I, Jansen AN, Abraham DP, Knuth J, Jones SA, Battaglia VS, et al. Differential voltage analyses of high-power, lithium-ion cells: 1. technique and application. *J Power Sources* 2005;139(1):295–303.
- [38] Bloom I, Walker LK, Basco JK, Abraham DP, Christophersen JP, Ho CD.

- Differential voltage analyses of high-power lithium-ion cells. 4. cells containing nmc. *J Power Sources* 2010;195(3):877–82.
- [39] Zheng G, Lee SW, Liang Z, Lee H-W, Yan K, Yao H, et al. Interconnected hollow carbon nanospheres for stable lithium metal anodes. *Nat Nanotechnol* 2014;9(8):618–23.
- [40] Watanabe S, Kinoshita M, Hosokawa T, Morigaki K, Nakura K. Capacity fade of LiAl_2O_3 cathode for lithium-ion batteries during accelerated calendar and cycle life tests (surface analysis of LiAl_2O_3 cathode after cycle tests in restricted depth of discharge ranges). *J Power Sources* 2014;258:210–7.
- [41] MATLAB, version 9.0.0.341360 (R2016a). Natick, Massachusetts: The Math-Works Inc.; 2016.
- [42] Söderström T, Stoica P, editors. *System identification*. Upper Saddle River, NJ, USA: Prentice-Hall, Inc.; 1988.